

Opinion

DNA methylation: unifying framework for complex chronic conditions

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Complex chronic conditions represent a growing health burden that is poorly understood and inadequately managed within current health systems. This review focuses on the potential of epigenetic DNA methylation to provide a pathway for better understanding of four closely related conditions: myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), long COVID (LC), fibromyalgia (FM), and hypermobile Ehlers–Danlos syndrome (hEDS). These conditions have extensive overlapping symptoms, suggesting shared underlying pathophysiological mechanisms. Epigenetic regulation, particularly DNA methylation, provides a powerful framework for comparative and longitudinal studies across these clinically overlapping conditions. Here, we synthesise evidence from current DNA methylation studies in ME/CFS, LC, and FM, and consider how an integrated cross-condition approach could advance mechanistic insight, enable diagnostic stratification, and improve patient outcomes.

Complex chronic conditions

Complex chronic conditions are ongoing, lifelong diseases that are currently poorly managed within health systems. The COVID-19 pandemic has added 400 million more cases to the health burden of countries worldwide [1]. This has heightened interest among clinicians and scientists in the frequently misunderstood related conditions. Much greater attention from the research and clinical communities is timely to highlight strategies that promise better cross-condition management and meaningful, evidence-based, effective treatments.

Currently, myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) [2], long COVID (LC) [3], and fibromyalgia (FM) [4] are stand-alone complex chronic conditions, and yet they share many overlapping symptoms, with multilayered molecular and physiological dysfunction spanning diverse biological systems [5]. Hypermobile Ehlers–Danlos syndrome (hEDS) is less well characterised but is a future comparative arm [6], as hypermobility is found in ME/CFS patients as well as in stand-alone hEDS. Consequential comorbidities, postural orthostatic tachycardia syndrome (POTS) [7] and mast cell activation syndrome (MCAS) [8], occur at significant frequencies in patients with these chronic conditions [5]. From the same triggering event, ME/CFS and FM can develop in different people [9,10]. A more coordinated, integrated approach to understanding the pathophysiology of these chronic conditions may provide cross-benefits to patients with each condition.

The landscape of each chronic condition**Myalgic encephalomyelitis/chronic fatigue syndrome**

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) describes post-viral or stressor-related debilitating conditions [11,12].¹ The name ‘ME’ was derived from an infectious disease outbreak in 1955 in London [13] and ‘CFS’ from an outbreak in 1984 in Incline Village, Nevada [14,15]. Geographically distinct ‘boutique’ infectious disease outbreaks, together with endemic

Highlights

Myalgic encephalomyelitis/chronic fatigue syndrome, long COVID, fibromyalgia, and hypermobile Ehlers–Danlos syndrome share overlapping clinical features and may involve common biological mechanisms.

DNA methylation provides a unifying epigenetic framework to investigate shared and disease-specific molecular signatures across these complex chronic conditions.

Recent methylome studies implicate immune, inflammatory, metabolic, and autonomic regulatory pathways in disease pathophysiology.

Comparative harmonised epigenetic analyses may enable patient stratification, identification of disease subtypes, and improved mechanistic understanding.

DNA methylation signatures have strong potential for developing early diagnostic tools and guiding precision medicine approaches for treatment options.

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infections from viruses such as Epstein–Barr virus, influenza, and enteroviruses [16], constantly give rise to ME/CFS. Nonviral triggers of severe stress also constantly drip-feed new cases, now estimated to be approximately 70 million worldwide [17].

The familial, intergenerational occurrence of ME/CFS suggested genetic linkage to the condition [18], and possible risk loci were highlighted in early genome-wide association studies (GWAS) [19,20]. Precision Life Oxford added a combinatorial platform to GWAS, so that clusters of single-nucleotide polymorphisms (SNPs) could be interrogated. Remarkably, 199 SNPs within 14 clusters, linked to just 15 genes, identified 91% of the ME/CFS samples in UK Biobank [21]. This indicated that a complex pattern of natural genetic variations, rather than distinct mutations, was responsible for susceptible individuals developing ME/CFS. In 2025, the DecodeME GWAS study of >15 000 ME/CFS patients revealed genetic linkage to a further 8 loci [22]. Application of the combinatorial platform analysis to the DecodeME dataset expanded the key SNPs to >7500, now linked to 259 genes [23], highlighting the complexity of genetic susceptibility risk factors.

Long COVID

LC emerged rapidly in large numbers as a post-viral syndrome from the COVID-19 pandemic [5,24,25]. Persistent clinical symptoms were given the name post-acute sequelae of SARS-CoV-2 (PASC) [3,26]. The World Health Organisation (WHO) proposed the name ‘Post-COVID Condition’ for symptoms persisting for 3 months. ⁱⁱ The popularised name LC, encompassing both those with an ME/CFS-like syndrome [27] and those with organ damage, is now used almost exclusively for the documented 65 million cases ⁱⁱⁱ and, more likely, 400 million cases [1]. Each new infection/reinfection with SARS-CoV-2, now an endemic virus, carries a risk of developing persistent, debilitating symptoms characteristic of LC [28]. Those who have recovered from LC within 2 years carry an increased risk of cardiovascular, neurological, and metabolic disorders [24]. With individual testing for COVID no longer occurring, there is an urgent need for new, effective diagnostic tools to identify and follow the progression of LC. Better treatment strategies would mitigate the long-term consequences for the many millions affected by LC and ME/CFS worldwide.

Genetic susceptibility derived from a combinatorial GWAS of LC patients identified 39 SNPs that could be linked to nine genes identified in the combinatorial analysis of ME/CFS patients [29]. A GWAS study with >6000 patients with LC highlighted the gene encoding the FOXP4 transcription factor, which is involved in lung physiology [30].

Fibromyalgia

FM is a complex chronic disorder affecting up to 5% of the global population. It is classified as a musculoskeletal disorder in which the characteristic feature is widespread body pain [31]. The increased sensitivity to pain is likely the result of the central nervous system not processing pain signals normally and muscle stiffness [4]. It has extensive symptom overlap with ME/CFS and LC, including fatigue, unrefreshing sleep, cognitive difficulties with memory loss and ‘brain fog’, as well as digestive problems similar to those seen in ME/CFS. Familial linkage in FM results in an 8- to 14-fold increase in the risk of developing the condition among first-degree relatives [32]. A recent 2026 GWAS study with 54 000 patients identified 26 loci linked to FM with strong neural connections [33].

Hypermobile Ehlers–Danlos syndrome

hEDS is the major subtype (80–90%) among the 13 connective tissue disorders arising from abnormal collagen formation and function [6]. It affects multiple organ systems and presents with clinical symptoms of skin hyperelasticity, joint hypermobility, and blood vessel fragility [34]. The genetic linkage of this subtype involves a combination of SNPs, as in ME/CFS and LC [35]. Despite hEDS being classified as a distinct condition, it manifests many of the same symptoms

as ME/CFS, LC, and FM, and up to 20% of patients diagnosed with ME/CFS also have hEDS, with 50% exhibiting joint hypermobility [36]. The genetic variations identified as causing hEDS are involved in the human leukocyte antigen (HLA) adaptive immune axis, the mitochondrial respiratory chain, and the collagen biosynthesis pathway, and therefore it could be classified as a particular subgroup of ME/CFS [37].

Shared symptoms and overlapping clinical features

The core overlapping symptoms for ME/CFS, LC, FM, and hEDS are fatigue, cognitive impairment, unrefreshing sleep, and generalised and specific muscle and joint pain. Post-exertional malaise (PEM), a response to stress or activity, is a hallmark symptom of ME/CFS but has now been well-documented in LC [38,39]. While FM patients have relapsing 'flares' that have not generally been accepted as PEM, recent publications have strongly advocated that PEM is a core symptom of FM [40,41]. The significant incidence of hypermobility in patients with ME/CFS complicates the understanding of whether PEM is a part of hEDS alone, but exercise intolerance has been reported [42]. Whether the similar FM flares and hEDS exercise intolerance are mediated by the same molecular and physiological changes as those in ME/CFS [43] needs to be determined and adds weight to the value of a more integrated approach. Dysautonomia is a feature of all conditions in which the regulation of involuntary physiological functions is abnormal, affecting blood pressure, cardiac rhythms, temperature, blood sugar, and gastrointestinal functions [44].

Key overlapping symptoms among the four chronic conditions, ME/CFS, LC, FM, and hEDS, are highlighted within the circle in Figure 1.

An epigenetic framework for complex chronic conditions

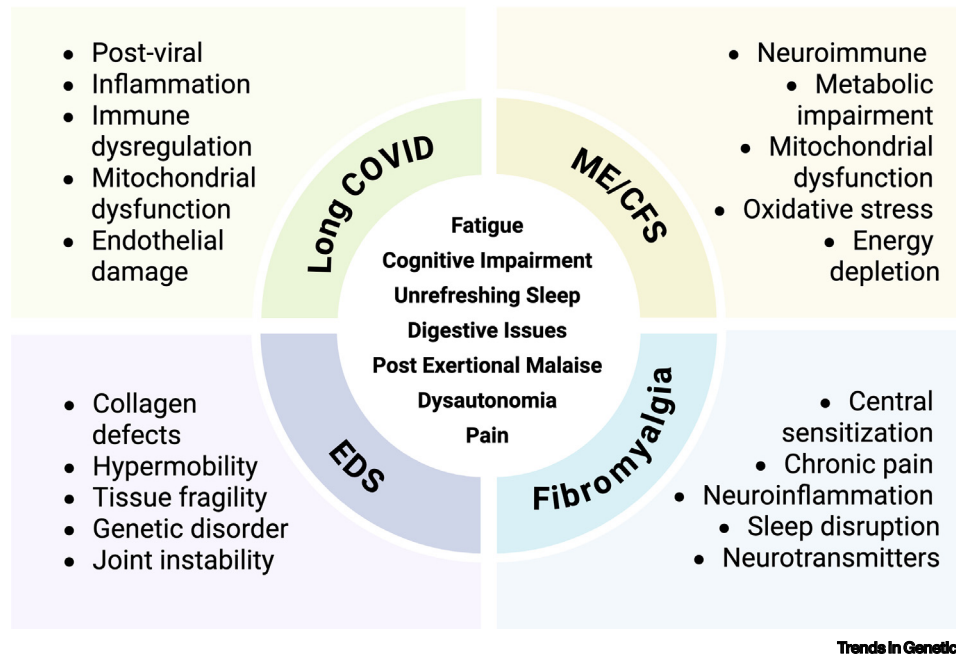
Various epigenetic characteristics have been investigated in complex chronic conditions: DNA methylation [45,46], histone modifications [47], and noncoding RNAs (ncRNAs) [48]. Epigenetic changes can occur as a result of external environmental influences such as stress, diet, exercise, and chemical pollutants that affect gene expression by modulating gene regulators [49].^{iv} Such changes can negatively impact human health, leading to disease [49,50].

DNA methylation provides a framework for investigating complex chronic conditions because it captures the dynamic interaction between genetic susceptibility, immune dysregulation, and persistent changes in gene regulation [51]. The DNA methylome profile can vary with the biological clock [52] and ageing [53], so it is important in patient studies of chronic diseases to use age- and sex-matched healthy controls. The authors have observed a very mild ageing effect in controls utilising principal component analysis, but not in the patient profiles, where the disease-related changes were dominant [54].

Multi-omics studies of ME/CFS have demonstrated that epigenetic DNA methylome analyses yielded stable, consistent changes across patients, compared with analyses of the transcriptome and proteome of peripheral blood mononuclear cells (PBMCs) [54–58]. The authors found that healthy controls had an approximately 0.1% change in DNA methylation month to month, whereas ME/CFS patients were more variable, at approximately 2% [58].

Comparative studies of ME/CFS and LC with standardised study parameters showed common methylation changes at cytosine-phosphate guanine (CpG) sites in both conditions [54]. CpG sites are regions of DNA where a cytosine nucleotide sits next to a guanine nucleotide along the 5' to 3' direction and are considered as the main target sites for DNA methylation. A minority of changes at CpG sites differed between the cohorts, although not among patients within the same cohort [54]. This implied that the selection of targeted regions could be suitable for

Overlapping Symptoms of the Chronic Diseases



Trends in Genetics

Figure 1. Comparative overview of shared and disease-specific symptoms across myalgic encephalomyelitis/chronic fatigue syndrome, long COVID, fibromyalgia (FM), and hypermobile Ehlers–Danlos syndrome (hEDS). Inner circle: core overlapping symptoms; outer blocks: evidence-based features often highlighted in the literature for each condition. Although PEM is less well defined in FM and hEDS, it is included here as a probable central overlapping symptom. CFS: chronic fatigue syndrome; ME: myalgic encephalomyelitis; PEM: post-exertional malaise;

diagnostic tests recognising both conditions and distinguishing between them. Dynamic epigenetic gene regulation at the onset of PEM, whether from mental stress (academic study) [58] or physical stress [cardiopulmonary exercise test (CPET) exercise test] [43], is associated with more severely dysfunctional physiological symptoms.

Understanding the pathophysiology of these complex conditions requires multiple technologies, as described in the recently published book on ME/CFS pathophysiology [59]. Since the conditions affect all body systems, future DNA methylation studies should interrogate multiple cell types in which changes to the methylomes are likely different. As shown in the following tables, a limited number of biological sources (whole blood, PBMCs, T-cell subtypes, and saliva) have been used. The authors are currently investigating plasma cell-free DNA (cfDNA), representative of DNA from cells throughout the body. Based on deconvolution of cell types in ME/CFS samples, cfDNA appears to be heavily dominated by DNA from blood cells. Single-cell DNA methylation services are commercially available and constitute a powerful tool at the research level to reveal cell-to-cell heterogeneity, both within the same cell type and among different cell types [60]. For patients generally to benefit, however, the preferred technology and approach must be translatable to clinical settings to be readily available to all clinicians and their patients.

ME/CFS, LC, and FM have more than 200 associated symptoms reported. Each patient has core symptoms and some of the other symptoms [61]. Hence, there is patient-to-patient heterogeneity that may inform classification into subgroups once more detailed molecular evidence is available. Improved systematic designs for comparative epigenetic DNA methylation analyses, using

matched biological fluids/tissues, age- and sex-matched patients and controls, and the same technologies and analytical frameworks, could provide critical new insights into shared and condition-specific mechanisms. Targeting common symptom domains within patient subgroups may help redress the lack of effective therapeutic strategies [62]. The findings of dynamic changes in methylation within longitudinal patient studies [43,58] highlight the potential of an epigenetic platform to facilitate strategies to reverse these changes [50].

Could a DNA methylation susceptibility signature be developed for these chronic conditions? A study by Jason and colleagues of 4500 University of Chicago students investigated blood samples taken prior to some contracting an Epstein–Barr virus infection. The preinfection blood samples of those who developed ME/CFS showed immune irregularities and lower cytokine levels, but not behavioural differences, compared with those who did not develop ME/CFS [63], as well as metabolic differences [64], implying a susceptibility ‘profile’ linked to development of the condition [65]. Such differences are inferred to be reflected in DNA methylation profiles.

How can epigenetic analysis close the knowledge gaps?

Major knowledge gaps greatly hinder our understanding and effective clinical management of chronic conditions. Epigenetic DNA methylation can be harnessed to address the lack of early biomarkers for molecular diagnostic tests for health management immediately after onset, rather than waiting several years for a definitive diagnosis [66]. As yet, possible molecular biomarkers for a simple, first-step early diagnosis that could be accessible to all clinicians and their patients have not been fully validated. ME/CFS, LC, FM, and hEDS all lack a simple molecular diagnostic test that would enable early diagnosis and better disease management, and at least a chance for partial recovery. Having a diagnostic platform with targeted, common methylation changes that could initially distinguish these chronic conditions from other diseases exhibiting persistent chronic fatigue would be a huge advance for patient management.

Tests based on epigenetic molecular signatures unique to each chronic condition are possible, as illustrated by methylation data from a study comparing ME/CFS and LC [54]. They could eliminate diagnostic ambiguity and enable appropriate best-practice management to begin as soon as possible. This first comparative study indicates that the DNA methylome has considerable promise for deriving molecular signatures with high selectivity and sensitivity for the development of definitive early diagnostic tests, as outlined in the strategy (Figure 2, key figure).

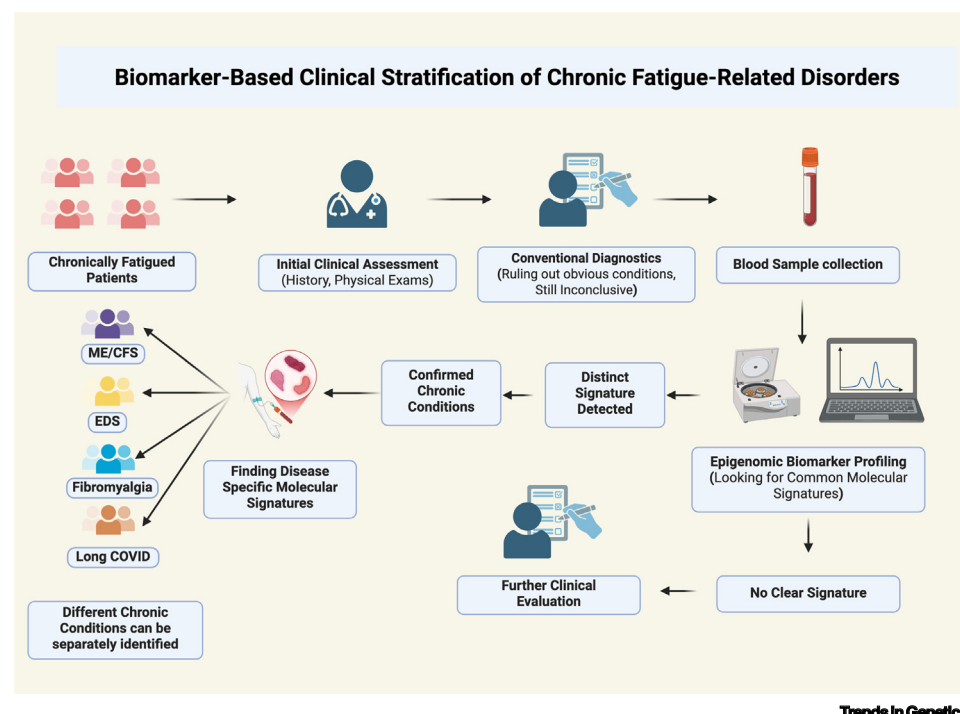
Advanced service platforms have been developed for assessing clinical epigenetic tests, including whole-genome bisulphite sequencing (WGBS) and chip technology (Infinium arrays) for measuring DNA methylation. These platforms have facilitated epigenomic diagnostics and the scientific validation of diagnostic tests for clinical use in genetic conditions. Genome-wide epigenetic signatures (episignatures) have been validated for diagnosis and phenotypic correlation in 42 Mendelian neurodevelopmental disorders, and recently episignatures have been investigated clinically using EpiSign METRIC (Methylation-based episignature testing and reference-based interpretation and classification) software [67–70].

Epigenetic studies on ME/CFS, LC, and FM

Here, we compare genome-wide DNA methylation studies across the three chronic conditions ME/CFS, LC, and FM (Table 1–3). The Ehlers–Danlos Society has signalled a study on epigenetic DNA methylation, ^v so hEDS could form a part of future comparative studies. A conference abstract has reported hEDS-specific DNA methylation episignatures in iPSC (induced pluripotent stem cell)-derived skin fibroblasts as a model for epigenetic dysregulation in hEDS pathogenesis [92].

Key figure

DNA methylation-based framework for early diagnosis of complex chronic conditions



Trends in Genetics

Figure 2. A proposed general strategy that utilise molecular signatures for diagnostic tests for chronic conditions based on the DNA methylome. CFS: chronic fatigue syndrome; EDS: Ehlers–Danlos syndrome; ME: myalgic encephalomyelitis.

DNA methylation studies on ME/CFS

Most DNA methylation studies have used Illumina arrays [71–76], and some have used reduced representation bisulphite sequencing (RRBS) [43,54,57,58,99]. Genome-wide studies in ME/CFS have primarily focused on PBMCs and specific immune cell subtypes (CD4+ and CD3+ T cells). Differentially methylated sites are typically defined using thresholds of approximately 5–15% change relative to healthy controls, although criteria have varied. The full body of published work from 2014 to 2026 is summarised in Table 1.

Array-based DNA methylome studies using Illumina 450K/EPIC platforms in PBMC-derived samples have identified widespread DMCs in ME/CFS. Most changes are hypermethylated (~72%) and associated with genes involved in cellular metabolism, immune regulation, and hypothalamic–pituitary–adrenal axis function [52,53]. A study of 70 female ME/CFS patients identified four distinct subtypes based on DNA methylation patterns [54]. Surprisingly, an Illumina 850K EPIC array-based study reported that approximately 98% of differentially methylated sites were *hypomethylated*, in sharp contrast to previous studies, but still predominantly associated with immune and cell signalling pathways [75].

Table 1. Comparisons of study features and significant observations of DNA methylation characteristics across the genome of ME/CFS patients

Year	Biological specimen.	Method	Diagnostic criteria	Cohort	Significant differences	Refs
2014	CD4+ T cells	Infinium Human Methylation 450K Bead Chip	Fukuda criteria	P = 25 (21 F and 4 M) C = 18 (10 F and 8 M)	120 CpGs (85% hypomethylated)	[71]
2014	PBMCs	Infinium Human Methylation 450K Bead Chip	Fukuda and Canadian criteria	P = 12, C = 12 All F	1192 CpGs (72% hypermethylated)	[72]
2017	PBMCs	Infinium Human Methylation 450K Bead Chip	Fukuda and Canadian criteria	P = 49 C = 25 All F	12 608 CpGs (72% hypermethylated)	[73]
2018	PBMCs	Infinium Human Methylation 450K Bead Chip	Fukuda and Canadian criteria	P = 70 All F	ME/CFS was divided into four different subtypes	[74]
2018	PBMCs	Illumina Methylation EPIC Microarray	Fukuda and Canadian criteria	P = 13 C = 12 All F	17 296 CpGs (98% hypomethylated)	[75]
2018	CD3+ T cells	Infinium Human Methylation 450K Bead Chip	Fukuda and Canadian criteria	P = 43 34 F and 9 M) C = 36 (27 F and 9 M)	133 DMPs (74% hypermethylated)	[76]
2020	PBMCs	Reduced representation bisulphite sequencing	Canadian criteria	P = 10 (5 F and 5 M) C = 10 (5 F and 5 M)	76 DMRs (52% hypomethylated) 394 CpGs (56% hypomethylated)	[57]
2022	PBMCs Longitudinal relapse study	Reduced representation bisulphite sequencing	Canadian criteria	P = 2, C = 1 All F	577 DMRs during relapse	[58]
2023	Meta-analysis of PBMCs and CD3+ T cells	Infinium Human Methylation 450K Bead Chip	Fukuda and Canadian criteria	PBMCs: P = 66 F C = 24 F CD3+ T cells: P = 61 (15 M and 46 F) C = 48 (12 M and 36 F)	CD3+ T cells: 510 CpGs (79% hypermethylated) PBMCs: 1751 CpGs (54% hypomethylated)	[99]
2025	PBMCs	Reduced representation bisulfite sequencing	Canadian	P = 5 (4 F and 1 M) C = 5 (4 F and 1 M)	214 DMRs (67% Hypermethylated) 118 DMRs common with LC DMRs	[54]
2025	PBMCs Longitudinal CPET PEM study	Reduced representation bisulphite sequencing	Canadian	P = 5 F, C = 2 F	205 DMRs 63 (0–24 h), 94(24–48 h) 120 (0–48 h)	[43]

C: control; CPET: cardiopulmonary exercise test; DMP: differentially methylated Probes; DMR: differentially methylated regions; F: female; M: male; P: patient; PBMC: peripheral blood mononuclear cells; PEM: post-exertional malaise.

More recent studies have used RRBS, a sequence-based approach for profiling genome-wide DNA methylation, for higher-resolution analysis of changes in ME/CFS (Table 1). By enriching for CpG-dense regulatory regions, RRBS enables more precise identification of differentially methylated regions (DMRs) and provides greater insight into their potential regulatory and functional significance [76]. The first RRBS-based methylation study in ME/CFS identified DMRs associated with 122 genes involved in neurological, immune, and metabolic pathways [57]. Subsequent longitudinal RRBS analysis examining PEM relapse–recovery dynamics in ME/CFS demonstrated that symptom exacerbation was accompanied by dynamic methylation changes at specific DMRs linked to inflammatory pathways, interleukin-8 signalling, nuclear factor

Table 2. Comparisons of study features and significant observations of DNA methylation characteristics across the genome of pre-LC and LC patients

Year	Time after infection	Biological specimen	Method	Cohort	Significant differences	Refs
2021	4 weeks post COVID-19 infection	Whole blood	Pyrosequencing	P = 117 C = 144	Acceleration of delta age (+5.25 above normal); Telomere shortening	[77]
2021	Few weeks post COVID-19 infection	PBMCs	Illumina Methylation EPIC 850K array	P = 38 (25 F and 13 M) C = 5	87 DMCs 66% hypermethylated	[78]
2022	3 months post COVID-19 infection	PBMCs	Whole genome bisulphite sequencing	P = 9, C = 5 All M	18 516 DMRs 53% hypermethylated	[79]
2022	3 months post COVID-19 infection	Whole blood	Infinium Human Methylation 450K Bead Chip	P = 109 (66 F and 43 M) C = 73 (50 F and 23 M)	168 DMRs 3 CpGs hypomethylated	[80]
2022	PASC symptoms for more than 12 weeks	PBMCs	Infinium Human Methylation 850K Bead Chip	P = 10 (7 F and 3 M) CC19 = 14 (7 F and 7 M) C = 17 (13 F and 4 M)	LC vs. CC19: 197 DMPs 65% hypermethylated LC vs Controls: 98 DMPs 57% hypermethylated	[81]
2022	1 years post COVID-19 infection	Circulating leukocytes	Infinium Human Methylation 850K Bead Chip	P = 15 (9 F and 6 M) C = 39 (21 F and 18 M)	71 DMRs 90% hypermethylated	[82]
2024	PASC symptoms for more than 4 weeks	Whole blood	Illumina NovaSeq 6000	P = 103 (80 F and 23 M) W = 15 (9 F and 6 M) C = 27 (14 F and 13 M)	39 DMR 87% hypermethylated	[83]
2024	6 months post COVID-19 infection	Whole blood	Infinium Human Methylation 850K Bead Chip	P = 96 (48 M and 48 F) C = 191 (141 M and 50 F)	42 DMCs 57% hypomethylated	[84]
2025	PASC symptoms for 3,6, and 12 months	PBMCs	Infinium Human Methylation 850K Bead Chip	P = 22 (18 M and 4 F) C = 22 (18 M and 4 F)	3 months vs healthy 16 611 DMCs 6 months vs healthy 6773 DMCs 12 months vs healthy: 4087 DMCs	[85]
2025	7 months post COVID-19 infection	Whole blood	Infinium MethylationEPIC BeadChip v1	P = 267 (137 M and 130 F) C = 79 (40 M and 39 F)	172 DMCs 73% hypermethylated	[86]
2025	1 years post COVID-19 Infection	PBMCs	Reduced representation bisulphite sequencing	P = 5 (4 F and 1 M) C = 5 (4 F and 1 M)	429 DMRs 66% hypermethylated, 118 DMRs common with ME/CFS	[54]

C: control; CC19: patients post COVID-19 with no PASC; DMC: differentially methylated cytosines; DMR: differentially methylated regions; P: patient; PBMC: peripheral blood mononuclear cells.

kappa-B (NFkB) dysregulation, and neutrophil degranulation. Notably, many of these methylation changes returned toward baseline during recovery, highlighting the dynamic and potentially reversible nature of epigenetic dysregulation in ME/CFS [58].

Table 3. Comparisons of study features and significant observations of DNA methylation characteristics of FM

Year	Biological specimen	Method	Cohort	Significant differences	Refs
2013	Whole blood	Illumina 450K HumanMethylation Chip	P = 10 C = 8	69 DMCs (91% hypermethylated)	[87]
2017	Whole blood	Illumina 450K HumanMethylation Chip	P = 24 C = 23 All F	1610 DMCs (65% hypomethylated)	[88]
2020	Leukocytes	Targeted next-generation bisulfite sequencing in 112 genomic regions	P = 8 All F C = 8 healthy sisters	2 DMRs 23 DMCs	[89]
2021	Leukocytes	Targeted next-generation bisulfite sequencing in 11 genomic regions	P = 42 All F C = 42 healthy sisters	1 DMR 1 DMC 17 DMCs without Bonferroni correction	[90]
2023	Meta-analysis Whole blood	Infinium Human Methylation 450K BeadChip; re-analysis with ChAMP	P = 22 C = 23 All F	1256 DMCs (75% hypomethylated)	[99]
2025	Saliva	Illumina Infinium MethylationEPIC array (850k BeadChip)	P = 53 78% F C = 1619 58% F	5 DMCs Hypermethylated	[91]

ChAMP: chip analysis methylation pipeline; DMC: differentially methylated cytosines; DMR: differentially methylated regions.

Further longitudinal RRBS analysis during a 48-h CPET protocol identified dynamic methylation changes associated with the onset of PEM, the hallmark symptom of ME/CFS. The study revealed distinct early (0–24 h) and late (24–48 h) DMRs at promoter-associated loci linked to endothelial function, inflammation, and immune regulation, uncovering epigenetic changes in stress-related immune functions. Remarkably, 98% of these methylation changes were not found in healthy controls, highlighting disease-specific epigenetic responses to physiological stress. Four distinct temporal methylation response patterns were identified, including reciprocal hypomethylation and hypermethylation dynamics across the 48-h period, further supporting the presence of dynamic and stress-responsive epigenetic dysregulation in ME/CFS [43].

DNA methylation studies in LC

DNA methylation studies on COVID-19 patients immediately postinfection have provided insight into the delay of the methylome to return to a pre-COVID state after viral infection, even when recovery was normal. These early studies, referred to here as pre-LC, focus on recovery and unresolved symptoms prior to the proposed diagnosis of LC 3 months after onset by the WHO. ^{vi} Studies with whole blood, PBMCs, and circulating leukocytes were carried out on pre-LC recovery/postinfection COVID patients, and LCs are described in Table 2.

Among studies with PBMCs, a pre-LC investigation using EPIC 850K arrays identified predominantly differentially hypermethylated CpGs (DMCs) affecting genes involved in immune modulation, inflammation, and viral response pathways [78]. Using the same technology with PBMCs from individuals who met the LC diagnostic criteria, DMCs were linked to prolonged activation of immune pathways, such as NFκB, and to disrupted immune modulation [79]. A study comparing LC patients with recovered individuals demonstrated a link to angiotensin II receptor, muscarinic receptor, and histamine signalling pathways [81]. By contrast, analyses

using peripheral blood rather than PBMCs highlighted DMCs associated with disturbed amino acid metabolism and Golgi apparatus functions [84].

The first longitudinal study of LC at 3, 6, and 12 months post-infection found a decrease in the number of DMCs over time compared with healthy controls. The *TXNRD1* gene, which encodes a protein that protects against oxidative stress, was highlighted as significantly associated with LC [85].

Studies on whole blood from COVID-19 survivors with pre-LC and LC conditions revealed biological age acceleration, telomere shortening, and modulation of ACE2 (Angiotensin-converting enzyme 2) expression, suggesting the occurrence of long-term epigenetic modifications that accelerate ageing and impair immune function [79]. Despite its relevance for the analysis of such post-viral conditions, whole blood has not yet been explored in genome-wide methylation studies of ME/CFS, limiting direct comparisons between the two conditions.

The one study to date that compared the DNA methylation changes between ME/CFS and LC using the same work stream [54] found that many of the 118 DMR changes in common were generally more pronounced in LC, perhaps reflecting time from onset (LC: 1 year vs. ME/CFS: 12 years). A small minority of DMRs were either absent in one condition or showed hypomethylation in one condition and hypermethylation in the other, thereby distinguishing the two conditions.

DNA methylation studies of fibromyalgia

DNA methylome studies in FM have used a greater diversity of tissue sources, including saliva and leukocytes. Both Illumina chip analyses and RRBS have been applied, with marked variation among the methodologies used.

A whole blood pilot study of women identified 69 predominantly hypermethylated CpG sites and increased chromosomal instability. The affected genes were linked to neuronal signaling, chromatin regulation, muscle function, oxidative stress, and inflammatory pathways [87].

A genome-wide whole blood study by contrast, identified 1,610 differentially methylated CpG sites, with most showing hypomethylation. These changes were enriched in pathways related to immune function, oxidative stress, DNA repair, and neuronal signalling, and were associated with altered cortical excitability [88]. The leukocyte studies with a large patient cohort produced a modest number of changes with the parameters used [89,90], and a saliva study revealed only five CpG differentially methylated sites, highlighting potential roles for immune and cellular signalling pathways in disease pathology [91].

Limitations of published DNA methylation studies

Despite limitations of heterogeneity in cohort size and composition, tissue sources, analytical platforms, and statistical thresholds, the studies reviewed here collectively have provided important insights into both the pathophysiology of the individual conditions and shared mechanisms. Heterogeneity and lifestyle confounders (smoking, BMI, medications, etc.) among patients with chronic conditions are potential limitations in currently published studies using this approach, but precision medicine studies of individual patients have shown that the dominant changes due to the condition override the influence of such confounders and do not materially affect the output data [54,57,58]. However, differences in experimental design and analytical approaches currently limit the validity of direct cross-study comparisons and hinder the identification of reproducible and more precise disease-associated methylation signatures. Future progress will require

greater standardisation of study design and epigenetic workflows, including harmonised patient phenotyping, longitudinal clinical sampling, tissue selection, sequencing methodologies, bioinformatic pipelines, and statistical criteria for defining differential methylation [93]. Standardised approaches will be particularly important for enabling robust cross-condition meta-analyses and for distinguishing shared epigenetic signatures from those arising from disease-specific molecular changes. Future studies should prioritise larger, clinically well-characterised cohorts, together with integration of multi-omic datasets, including transcriptomic, proteomic, metabolomic, and immunological profiling, to better define the biological networks underlying these conditions [94].

Importantly, functional validation is also essential to establish whether observed methylation changes are causative, compensatory, or secondary consequences of chronic disease processes [95]. Experimental studies using cellular systems, single cells, immune models, organoids, and locus-specific epigenetic editing approaches will help determine the regulatory

Pathways in ME/CFS

Immune dysregulation — T-cell receptor signalling and activation, immune regulation, primary immunodeficiency, MHC-associated processes, leukocyte differentiation, and immune-response regulation.

Inflammatory and stress signalling — IL-1, IL-8, IL-10, IL-17A, p38 MAPK/JNK signalling, response to wounding, and cellular stress-response pathways.

Metabolic and energy regulation — ATP/nucleotide and phosphate metabolism, AMPK signalling, PPAR α /RXR α activation, kinase activity, and metabolic pathways.

Neuroendocrine and neurological regulation — GABAergic, glutamatergic, serotonergic and dopaminergic signalling, calcium/cAMP signalling, neuroendocrine signalling, circadian regulation, neuronal and nerve development.

Cellular regulation and survival — apoptosis/proliferation, transcriptional regulation, Wnt signalling, tyrosine-kinase signalling, and developmental regulation.

Cytoskeletal regulation — cytoskeletal remodelling, cell adhesion/migration, and Rho GTPase signalling.

Pathways in Long COVID

Immune dysregulation — regulation of immune effector processes, mast cell activation, interferon- α production, lymphocyte differentiation, immune-response-regulating signalling, adaptive immune-system pathways, neutrophil degranulation, and defence response to virus.

Inflammatory and vascular signalling — integrin signalling, cell adhesion, haemostasis, angiotensin II signalling, histamine H1 receptor signalling, and VEGF signalling.

Metabolic and intracellular signalling — metabolism of lipids, nucleoside and nucleotide binding, Rho GTPase signalling, receptor tyrosine kinase signalling, G-protein signalling, PI3K signalling, and Wnt signalling.

Cytoskeletal regulation — actin-cytoskeleton pathways, integrin signalling, cell adhesion, and Rho GTPase signalling.

Neuroendocrine and neurological regulation — circadian-rhythm pathways, muscarinic acetylcholine receptor 1 and 3 signalling, histamine H1 receptor signalling, and neuronal development.

Cellular regulation and survival — apoptosis signalling, Wnt signalling, receptor tyrosine kinase signalling, and PI3K signalling.

Pathways in Fibromyalgia

Neurological development and neuronal regulation — neuron differentiation, nervous-system development, axonal bundling and outgrowth, neuronal excitability and synaptic transmission, pain regulation, and neuroendocrine membrane trafficking.

Developmental and structural processes — anatomical, system, organ and multicellular-organism development, skeletal-system development, fibronectin-related processes, and cell adhesion.

Muscle and neuromuscular regulation — muscle contraction, muscle-cell signalling, and muscle maturation.

Immune, inflammatory and cellular stress responses — immune regulation, inflammatory processes, oxidative-stress responses, and DNA-damage/repair processes.

Metabolic, mitochondrial and intracellular regulation — protein metabolism, mitochondrial translation and translational elongation/termination, autophosphorylation, GPCR signalling, calcium-ion binding, glycosylation, and glycoprotein-related processes.

Circadian regulation — circadian sleep/wake-cycle processes and regulation of wakefulness.

Transcriptional and chromatin regulation — regulation of transcription elongation, chromatin compaction, chromosomal segregation, and related cellular regulatory processes.

Common Pathways

- Inflammatory signalling
- Immune dysregulation
- Metabolic dysfunction/regulation
- Neurological and neuronal-development pathways
- Intracellular signal transduction
- Actin-cytoskeleton pathway
- Cellular stress
- Circadian regulation

Trends in Genetics

Figure 3. Biochemical pathways and physiology affected by DNA methylation changes in studies of ME/CFS, LC, and FM. Information from the gene enrichment analyses of the published studies shown in Tables 1–3 was collated and is illustrated, for each of ME/CFS, LC, and FM individually, in the figure, together with the altered physiology common to all conditions. For the individual list, please refer to Supplementary Table S1 in the Supplemental information online. AMPK: 5' adenosine monophosphate-activated protein kinase; cAMP: cyclic adenosine monophosphate; FM: fibromyalgia; GPCR: G protein-coupled receptor; JNK: c-Jun N-terminal kinase; LC: Long COVID; MAPK: mitogen-activated protein kinase; MHC: major histocompatibility complex; PI3K: phosphoinositide 3-kinase; PPAR: peroxisome proliferator-activated receptor; RXR: retinoid X receptor; VEGF: vascular endothelial growth factor.

impact of candidate methylation changes on gene expression and downstream biological pathways. Such mechanistic studies will be critical for translating epigenetic discoveries into clinically meaningful biomarkers and future targeted therapeutic strategies for complex chronic conditions.

Improving treatment options for complex chronic conditions

Despite the substantial clinical burden, effective evidence-based therapies for these conditions remain extremely limited, highlighting the urgent need for biologically informed precision medicine approaches [62]. Changes in DNA methylation patterns can identify promising targets within biochemical pathways for the development of evidence-based, specific, targeted treatments. Since effective treatments are so limited, a new paradigm is needed, tailored to the complexity of these chronic conditions, rather than targeting individual symptoms. Despite the limitations due to extensive heterogeneity in the analytical parameters used in the studies in Tables 1–3 and Table S1 in the Supplemental information online, combined gene enrichment analysis of the individual studies (Figure 3) reveals key regulatory pathways and physiological processes for targeted treatments for ME/CFS, LC, and FM, as well as key overlapping physiological aspects that could inform common treatments.

Notably, many studies consistently identified dysfunctions in inflammatory signalling, immune regulation, metabolic pathways, neurological function, intracellular signalling, the cytoskeleton, and cellular stress, supporting the concept that these clinically overlapping conditions may share common underlying pathophysiological processes that could inform more targeted treatments.

The conditions ME/CFS, LC, and FM have traditionally been thought of, observationally, as fluctuating but not progressively neurodegenerative like Parkinson's or Alzheimer's disease;

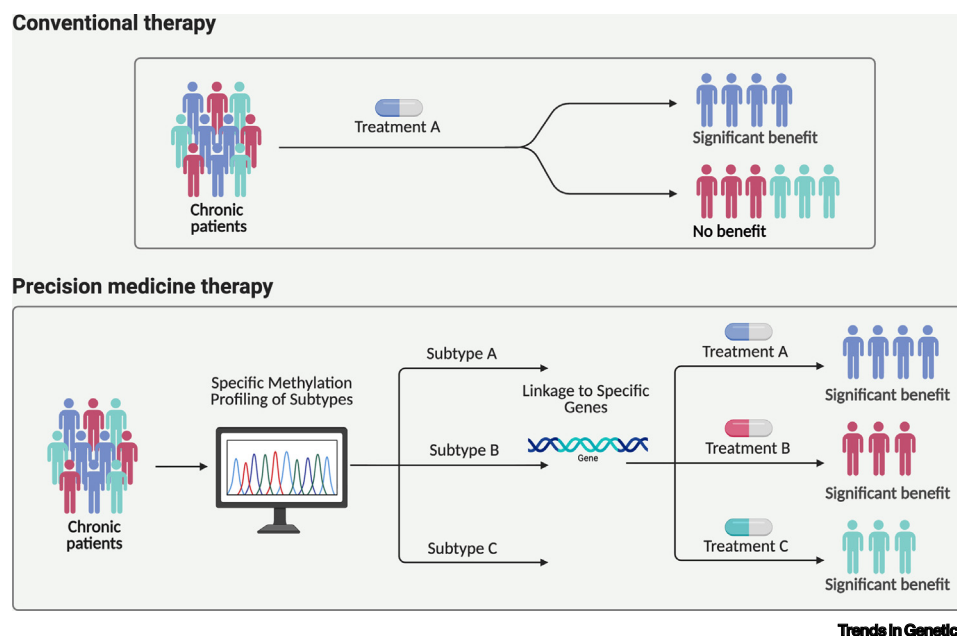


Figure 4. A role for DNA methylation in identifying subtypes and advancing treatments for ME/CFS, LC, FM, and EDS. The schematic diagram illustrates the potential use of DNA methylation profiling to stratify heterogeneous populations into molecular subtypes. Linking subtype methylation signatures to genes and biological pathways may help inform the selection of more targeted therapies, potentially improving treatment response compared with trial-and-error approaches. CFS: chronic fatigue syndrome; EDS: Ehlers–Danlos syndrome; FM: fibromyalgia; LC: Long COVID; ME: myalgic encephalomyelitis.

however, there is new evidence of neuronal loss from the first ME/CFS autopsy studies ^{vii} and there may be features that could be classified as neurodegenerative [96]. However, there are also documented cases of complete recovery to normal health [97], ^{viii} implying that the possibility of finding a mechanism or ‘switch’ to reverse the conditions remains a key goal to relieve the growing health burden on patients and societal health systems.

Collectively, these observations support a shift towards longitudinal, precision medicine approaches that track individual patients over time to better define treatment pathways. Indeed, studies of clinically well-characterised patients have shown that small cohorts can yield highly informative conclusions about their individual pathophysiology and provide insights into variations among individual patients [54]. Patients with these chronic conditions will likely be grouped into particular subtypes, reflecting their responses to treatments that are linked to specific changes in molecular biology and physiology. Once their subtype is determined, more targeted treatments could be derived (as illustrated in the schematic in Figure 4), rather than the current ineffective ‘trial-and-error’ approaches.

Concluding remarks and future perspectives

The global emergence of Long COVID has brought unprecedented visibility to complex chronic conditions, transforming what were once ignored or poorly understood conditions into a major public health priority. This has exposed critical gaps in healthcare systems, with delayed diagnosis, limited therapeutic options, and fragmented models of care. These observations explain the longstanding challenges faced by patients and their absence from mainstream clinical response pathways, leading them to rely instead on patient support networks with limited resources. The scale and urgency of LC now provide a unique opportunity and impetus to evaluate and strengthen clinical, research, and policy frameworks for this broader class of chronic conditions. For LC, a relatively recent condition existing for 5–6 years, insights from patients with ME/CFS who have been affected for much longer periods since onset can provide insights into the ongoing LC pathophysiology [98].

Addressing these challenges will require new approaches that capture the biological complexity underlying these conditions (see Outstanding questions). Converging evidence of multilayered dysregulation across immune, metabolic, neurological, and autonomic systems indicates that single-symptom/discipline-focused models are inadequate. As described in this article, epigenetic regulation, particularly DNA methylation, offers a powerful framework for linking environmental triggers, host susceptibility, and persistent changes in gene regulation. Its relative stability, sensitivity, and accessibility in peripheral tissues position it as a leading candidate for biomarker development, disease stratification, and monitoring of disease trajectories. Future approaches, however, should also prioritise the integration of multi-omic data and well-characterised clinical cohorts to disentangle the shared and condition-specific mechanisms. Rather than representing entirely distinct disorders, ME/CFS, LC, FM, and EDS may ultimately be shown to exist along an overlapping epigenetic and physiological spectrum, shaped by shared mechanisms of maladaptive chronic stress responses.

Translating the current biological and epigenetic advances into clinical practice will require consistently applied, robust, scalable diagnostic tools that can be implemented in routine healthcare settings. The identification of shared and condition-specific epigenetic signatures could enable a tiered diagnostic approach to distinguish specific disease subtypes. Such stratification would provide a critical foundation, allowing treatments to be tailored to the underlying molecular and physiological profiles of individual patients, rather than relying solely on current trial-and-error approaches.

Outstanding questions

To what extent do myalgic encephalomyelitis /chronic fatigue syndrome, long COVID, fibromyalgia, and hypermobile Ehlers–Danlos syndrome share common DNA methylation signatures that reflect convergent biological mechanisms?

Do distinct epigenetic subtypes exist within each condition, and could these support precision medicine approaches?

How closely do methylation changes correspond with changes in gene expression, protein abundance, and physiological dysfunction?

Which tissues and cell populations provide the most informative methylation signatures for diagnosis and mechanistic studies?

Are methylation signatures different in the cell types being analysed?

Which methylation changes are disease-specific, and can they reliably distinguish related chronic conditions from one another?

Are observed DNA methylation changes causal drivers of disease processes, compensatory responses, or downstream consequences of chronic illness?

Can DNA methylation biomarkers be translated into robust, cost-effective diagnostic tests suitable for routine clinical practice?

Could therapeutic interventions that target epigenetically dysregulated pathways reverse disease-associated molecular changes and improve patient outcomes?

Such an aspirational landscape will enable timely diagnosis and effective intervention to achieve an improved quality of life for millions of individuals affected by complex chronic conditions worldwide.

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Declaration of interests

The authors declare no competing interests.

Resources

- ⁱ<https://www.cdc.gov/me-cfs/hcp/diagnosis/iom-2015-diagnostic-criteria-1.html>
- ⁱⁱhttps://www.who.int/publications/i/item/WHO-2019-nCoV-Post_COVID-19_condition-Clinical_case_definition-2021.1
- ⁱⁱⁱ<https://data.who.int/dashboards/covid19/cases>
- ^{iv}<https://www.encyclopédie-environnement.org/en/health/epigenetics-how-the-environment-influences-our-genes-2/>
- ^v<https://www.ehlers-danlos.com/heds-hsd-research/>
- ^{vi}https://www.who.int/publications/i/item/WHO-2019-nCoV-Post_COVID-19_condition-Clinical_case_definition-2021.1
- ^{vii}<https://www.healthrising.org/blog/2025/12/04/chronic-fatigue-hpa-axis/>
- ^{viii}https://www.healthrising.org/blog/2020/01/05/christines-dnrs-chronic-fatigue-recovery-story/?utm_source=chatgpt.com

Supplemental Information

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